

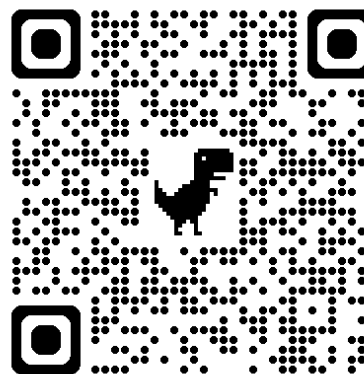


# 自然語言處理與應用

# Natural Language Processing and Applications

## LLM Evaluations

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2026/05/11

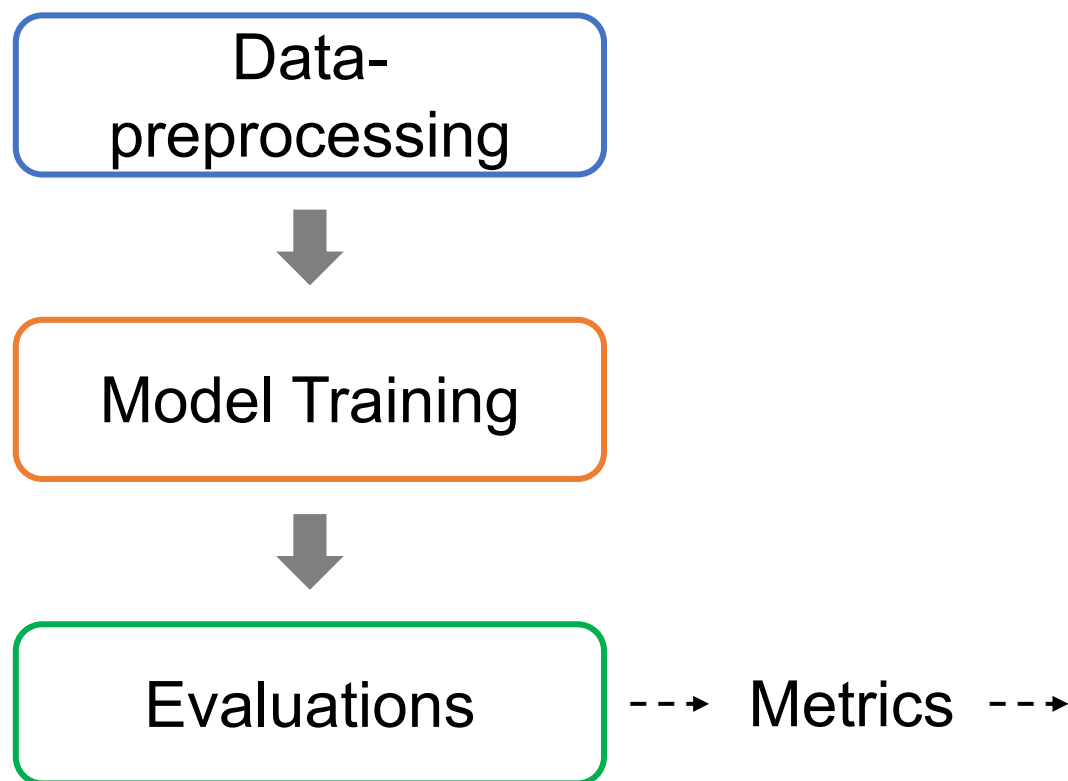


GitHub



Slido  
# NLP0511  
([Link](#))

# Phases for Building a Machine Learning Model

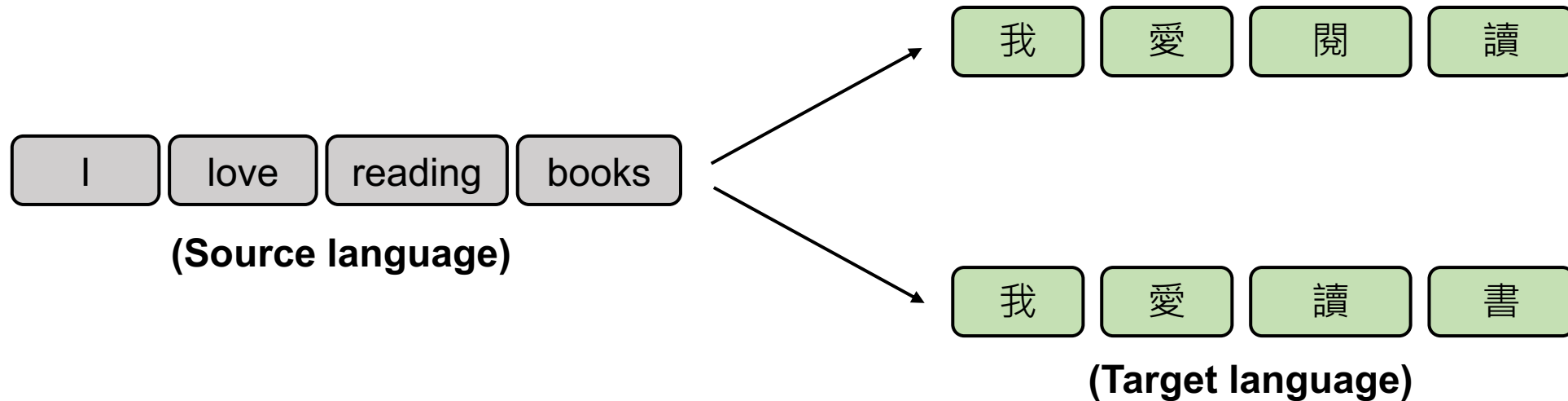


# How to evaluate natural language generation?

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- Natural language is hard to evaluate due to subjectivity and language diversity.

For example: Machine Translation



- Human evaluations
- Automatic evaluations (We will focus on this topic.)

# Medical Report Example

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**Ground Truth Report:** the **lungs** are hyperexpanded consistent with **emphysema** the heart size and pulmonary vascularity appear within normal limits no **pneumothorax** or **pleural effusion** is seen patchy **airspace** disease is present in the right middle lobe degenerative changes are present **spine**

## Model Generated report

- There is hyperexpansion of the **lungs** indicating **emphysema**.
- The heart and pulmonary vessels are within normal limits.
- No signs of **pneumothorax** or **pleural effusion**.
- **Airspace** disease affects the right middle lobe.
- Degenerative spinal changes are noted.

**Is this good?**

# Evaluations

- Perplexity
- BLEU
- ROUGE

Rule-based  
(model-free)

- BERTScore
- BLEURT

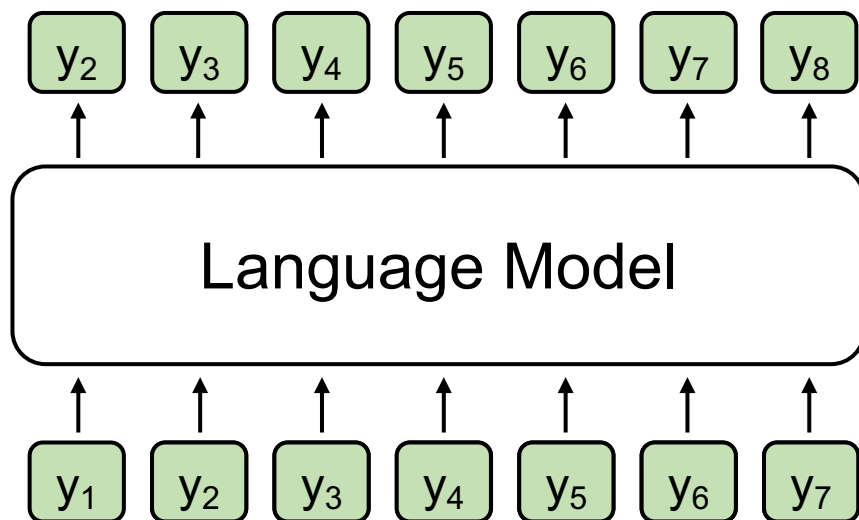
Learning-based  
(model-based)

- GPTScore

LLM-based

# [Recap] 訓練 GPT 最大化每個時間點的機率

$$P(y_2|y_1) P(y_3|y_1, y_2) \quad \dots \quad P(y_8|y_1, y_2, y_3, y_4, y_5, y_6, y_7)$$



目標函數：

$$\prod_{t=1}^n P(y_t | y_1, y_2, \dots, y_{t-1}) \quad \leftarrow \text{Language Modeling}$$

 Self-attention 不可及的範圍

# [Recap] Language Modeling and Cross-entropy

---

整句機率是很多條件機率一路相乘 (隨著 $t$ 增加結果接近 0)，因此通常會取 $\log$ ：

$$\log\left(\prod_{t=1}^n P(y_t | y_1, y_2, \dots, y_{t-1})\right)$$

$$= \sum_{t=1}^n \log P(y_t | y_1, y_2, \dots, y_{t-1}) \quad \leftarrow \text{越大越好}$$

加上負號之後就可以計算 **Cross-entropy**  
(**loss** 越小越好)

# Perplexity (PP)

<https://zh.wikipedia.org/zh-tw/幾何平均數>

- Perplexity 定義：entropy 的指數形式  $PP(p) = b^{H(p)}$
- 直觀理解：language modeling 放在分母的幾何平均數
  - Why 放在分母？主義是定義問題->把 entropy 從「bits」這個抽象單位變成「有效選項數」這個直覺單位

在數學中，幾何平均數是一種**均值**，它通過使用它們的值的乘積（**算術平均數**使用"和"）來指示一組數字的集中趨勢或典型值。幾何平均數定義為第 $n$ 根個數的**乘積**的第 $n$ 個根，即對於一組數字 $x_1, x_2, \dots, x_n$ ，幾何平均數定義為：

$$\left( \prod_{i=1}^n x_i \right)^{\frac{1}{n}} = \sqrt[n]{x_1 x_2 \cdots x_n}$$

**Perplexity:**

$$\left( \prod_{t=1}^n \frac{1}{P(y_t | y_1, y_2, \dots, y_{t-1})} \right)^{\frac{1}{n}}$$



# Perplexity 簡單推導

---

$$P(y_t|y_1, y_2, \dots, y_{t-1}) = P(y_t|y_{<t})$$

$$H(p) = -\frac{1}{n} \sum_{t=1}^n \log_b P(y_t|y_{<t}) = -\frac{1}{n} \log_b \prod_{t=1}^n P(y_t|y_{<t})$$

$$PP(p) = b^{H(p)} = b^{-\frac{1}{n} \log_b \prod_{t=1}^n P(y_t|y_{<t})} = b^{\log_b \prod_{t=1}^n P(y_t|y_{<t})^{-\frac{1}{n}}} = \prod_{t=1}^n P(y_t|y_{<t})^{-\frac{1}{n}}$$

# Perplexity and geometric mean

---

以生成10個tokens為例 (前9個機率值都是0.9，最後一個機率值0.1):

算數平均數： $(0.9 * 9 + 0.0001)/10 = 0.81001$

幾何平均數： $\sqrt[10]{0.9^9 * 0.0001} = 0.3621$

# Perplexity 的另外一種觀察

---

以生成3個tokens為例， $p_1 = \frac{1}{2}, p_2 = \frac{1}{4}, p_3 = \frac{1}{8}$

$$PP = 64^{\frac{1}{3}} = 4 = \left(\frac{1}{\frac{1}{4}}\right)^{\frac{1}{3}}$$

代表平均模型「在每一步都從 4 個等機率選項中亂猜並猜中」的難度

# BLEU (Bilingual Evaluation Understudy)

---

常用於機器翻譯

- A word-based metric.
  - It is very sensitive to word tokenization
- Core concept: Compute **precision** for n-grams:
  - Unigrams -> BLEU-1
  - Bigrams -> BLEU-2
  - Trigrams -> BLEU-3
  - 4-grams -> BLEU-4

# Precision and Recall (「檢索」角度的定義)

---

$$\text{Precision} = \frac{\text{Relevant and retrieved instances}}{\text{All retrieved instances}} \longleftarrow \text{Predicted by a model}$$

$$\text{Recall} = \frac{\text{Relevant and retrieved instances}}{\text{All relevant instances}} \longleftarrow \text{Ground-truths}$$

Relevant and retrieved instances: **Intersection** between predictions and ground-truths

# Calculation of BLEU Score (Example)

---

Assume we now translate from Chinese to English.

機器翻譯常用兩個  
以上 references

**Calculate BLEU-1 score**

Chinese: 我想要讀那本書

Precision:  $\frac{6}{6}$

Reference1: I want to read the book.

Reference2: I want to read that book.

Model output: the the the the the the.

100%! Can this be true?

# Calculation of BLEU Score (Example)

---

Assume we now translate from Chinese to English.

**Calculate BLEU-1 score**

Chinese: 我想要讀那本書

Reference1: I want to read the book.

Reference2: I want to read that book.

Model output: the the the the the the.

~~Precision:  $\frac{6}{6}$~~

Modified Precision:  $\frac{1}{6}$

# Why should we use modified precision?

---

- The output sequences can be total mistakes.
  - E.g., the the the the the the
- Original precision is in favor of longer output sequences.
- Therefore, we should use modified precision to prevent bad evaluations.



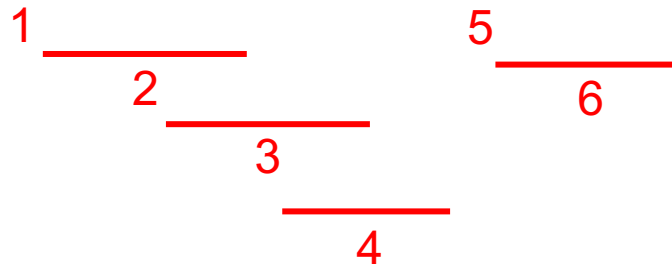
# Calculation of BLEU Score (Example)

## Calculate BLEU-2 score

Reference1: The dog is on the bed.

Reference2: There is a dog on the bed.

Model output: The dog the dog on the bed.



|         | Count |             |
|---------|-------|-------------|
| the dog | 2     | (uplicated) |
| dog the | 1     |             |
| dog on  | 1     |             |
| on the  | 1     |             |
| the bed | 1     |             |

# Calculation of BLEU Score (Example)

## Calculate BLEU-2 score

Reference1: The dog is on the bed.

Reference2: There is a dog on the bed.

Model output: The dog the dog on the bed.

|         | Count | Clips to the reference<br>↓<br>Count <sub>clip</sub> |
|---------|-------|------------------------------------------------------|
| the dog | 2     | 1                                                    |
| dog the | 1     |                                                      |
| dog on  | 1     |                                                      |
| on the  | 1     |                                                      |
| the bed | 1     |                                                      |

$$\text{Count}_{\text{clip}} = \min(\text{Count}, \text{Max\_Ref\_Count})$$

# Calculation of BLEU Score (Example)

---

## Calculate BLEU-2 score

Reference1: The dog is on the bed.

Reference2: There is a dog on the bed.

Model output: The dog the dog on the bed.

|         | Count | Count <sub>clip</sub> |
|---------|-------|-----------------------|
| the dog | 2     | 1                     |
| dog the | 1     | 0                     |
| dog on  | 1     |                       |
| on the  | 1     |                       |
| the bed | 1     |                       |

$$\text{Count}_{\text{clip}} = \min(\text{Count}, \text{Max\_Ref\_Count})$$

# Calculation of BLEU Score (Example)

---

## Calculate BLEU-2 score

Reference1: The dog is on the bed.

Reference2: There is a dog on the bed.

Model output: The dog the dog on the bed.

|         | Count | Count <sub>clip</sub> |
|---------|-------|-----------------------|
| the dog | 2     | 1                     |
| dog the | 1     | 0                     |
| dog on  | 1     | 1                     |
| on the  | 1     |                       |
| the bed | 1     |                       |

$$\text{Count}_{\text{clip}} = \min(\text{Count}, \text{Max\_Ref\_Count})$$

# Calculation of BLEU Score (Example)

---

## Calculate BLEU-2 score

Reference1: The dog is on the bed.

Reference2: There is a dog on the bed.

Model output: The dog the dog on the bed.

Count **only one time** even mapped to both references.

$\text{Count}_{\text{clip}} = \min(\text{Count}, \text{Max\_Ref\_Count})$

|         | Count | Count <sub>clip</sub> |
|---------|-------|-----------------------|
| the dog | 2     | 1                     |
| dog the | 1     | 0                     |
| dog on  | 1     | 1                     |
| on the  | 1     | 1                     |
| the bed | 1     |                       |

# Calculation of BLEU Score (Example)

## Calculate BLEU-2 score

Reference1: The dog is on the bed.

Reference2: There is a dog on the bed.

Model output: The dog the dog on the bed.

Count **only one time** even mapped to both references.

|         | Count | Count <sub>clip</sub> |
|---------|-------|-----------------------|
| the dog | 2     | 1                     |
| dog the | 1     | 0                     |
| dog on  | 1     | 1                     |
| on the  | 1     | 1                     |
| the bed | 1     | 1                     |

Modified Precision:  $\frac{4}{6}$

# Formula of BLEU Score

---

Summation for unigram, bigram, tri-gram, and 4-gram

$$p_n = \frac{\sum_{C \in \{Candidates\}} \sum_{n\text{-gram} \in C} Count_{clip}(n\text{-gram})}{\sum_{C' \in \{Candidates\}} \sum_{n\text{-gram}' \in C'} Count(n\text{-gram}')}$$

Summation for all candidates (model outputs) of each translation

# ROUGE Score

---

- ROUGE (Recall-Oriented Understudy for Gisting Evaluation)
  - Mainly for text summarization
- Metric Input: **Summary** (**prediction**), **Reference** (**gold summary**)
- Common metrics: ROUGE-1, ROUGE-2, ROUGE-L
  - L: Longest common subsequence
- **Please note that current papers calculate ROUGE-F as default!!!**
  - In other words, ROUGE-1**F**, ROUGE-2**F**, ROUGE-L**F**



# ROUGE-1 Example

---

predictions = ["The", "cat", "sat", "on", "the", "mat"]

references = ["A", "cat", "was", "sitting", "on", "the", "mat"]

ROUGE-1 **recall** = Number of matching unigrams / Number of unigrams in the reference =  $4/7$

ROUGE-1 **precision** = Number of matching unigrams / Number of unigrams in the machine-generated summary =  $4/6$

ROUGE-1 **F1-score** = Harmonic mean of the precision and the recall =  $2 * 4/7 * 4/6 / (4/7 + 4/6)$

# ROUGE-2 Example

---

predictions = ["The cat", "cat sat", "sat on", "on the", "the mat"]

references = ["A cat", "cat was", "was sitting", "sitting on", "on the", "the mat"]

ROUGE-2 **recall** = Number of matching bigrams / Number of bigrams in the reference =  $2/6$

ROUGE-2 **precision** = Number of matching bigrams / Number of bigrams in the machine-generated summary =  $2/5$

ROUGE-2 **F1-score** = Harmonic mean of the precision and the recall =  $2 * 2/6 * 2/5 / (2/6 + 2/5)$

# ROUGE-L Example

- L: Longest common subsequence

predictions = ["The", "cat", "sat", "on", "the", "mat"]

references = ["A", "cat", "was", "sitting", "on", "the", "mat"]

The longest common subsequence is ["cat", "on", "the", "mat"]

ROUGE-L **recall** = Number of matching unigrams / Number of unigrams in the reference = 4/7

ROUGE-L **precision** = Number of matching unigrams / Number of unigrams in the machine-generated summary = 4/6

ROUGE-L **F1-score** = Harmonic mean of the precision and the recall =  $2 * 4/7 * 4/6 / (4/7 + 4/6)$

# ROUGE-L Example

- L: Longest common subsequence

The order should be kept for the LCS problem

predictions = ["The", "cat", "sat", "on", "the", "mat"]

references = ["on", "the", "mat", "sitting", "a", "cat"]

The longest common subsequence is ["on", "the", "mat"]

ROUGE-L **recall** = Number of matching unigrams / Number of unigrams in the reference = 3/6

ROUGE-L **precision** = Number of matching unigrams / Number of unigrams in the machine-generated summary = 3/6

ROUGE-L **F1-score** = Harmonic mean of the precision and the recall =  $2 * 0.5 * 0.5 / (0.5 + 0.5)$

# Precision (BLEU) vs. Recall (ROUGE)?

---

- 翻譯有「標準答案」的感覺：好的翻譯應該只說該說的,多說的話就是亂加東西  
→ 在乎 **precision** (產出的內容裡有多少是對的)
- 摘要要有「抓重點」的感覺：好的摘要應該把重要的點都涵蓋到  
→ 在乎 **recall** (該說的內容有多少被說到)

# Issue of BLEU and ROUGE

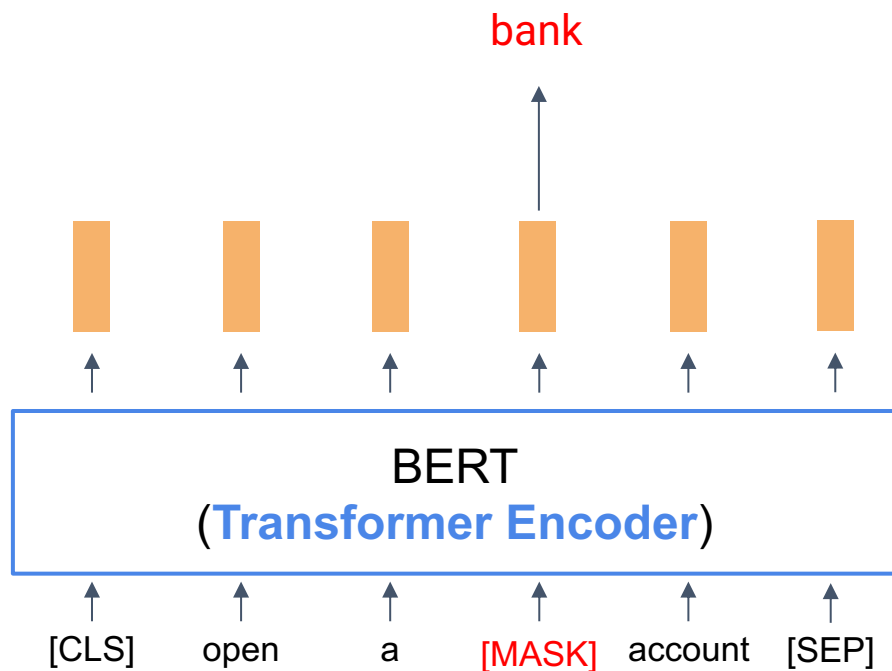
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- Pros: 採用 tokens 重疊率，容易實作
- Cons: Cannot meet language diversity
  - This mainly comes from the way for measuring **overlapping** rates.
- **Question:** **Can we create an automatic metric to fix the issue?**
- Next, we are going to introduce two **learned** automatic evaluation metrics
  - **BERT**Score (ICLR 2020)
  - BLEURT (ACL 2020)

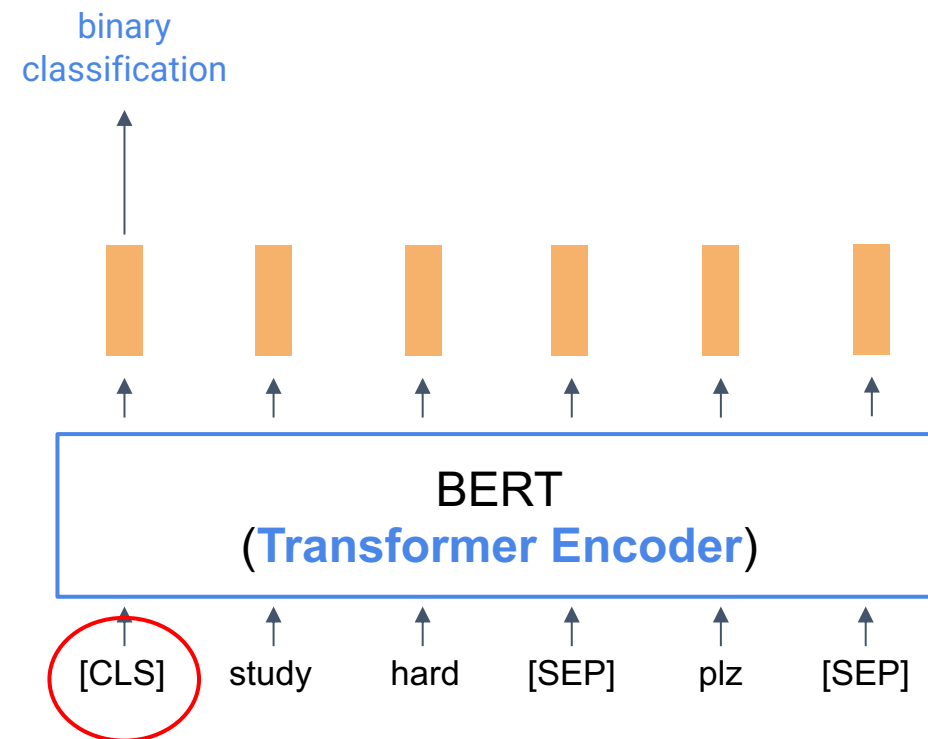
# [Recap] BERT (**Bidirectional** Encoder Representations from Transformers)

- BERT 有兩種預訓練任務：

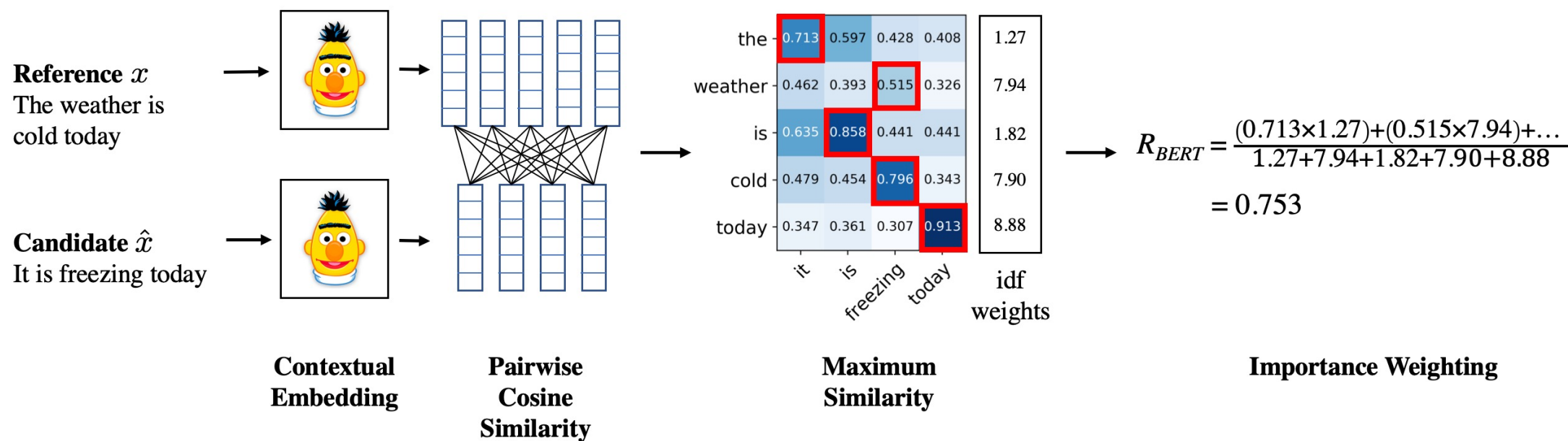
## Masked Language Modelling



## Next Sentence Prediction

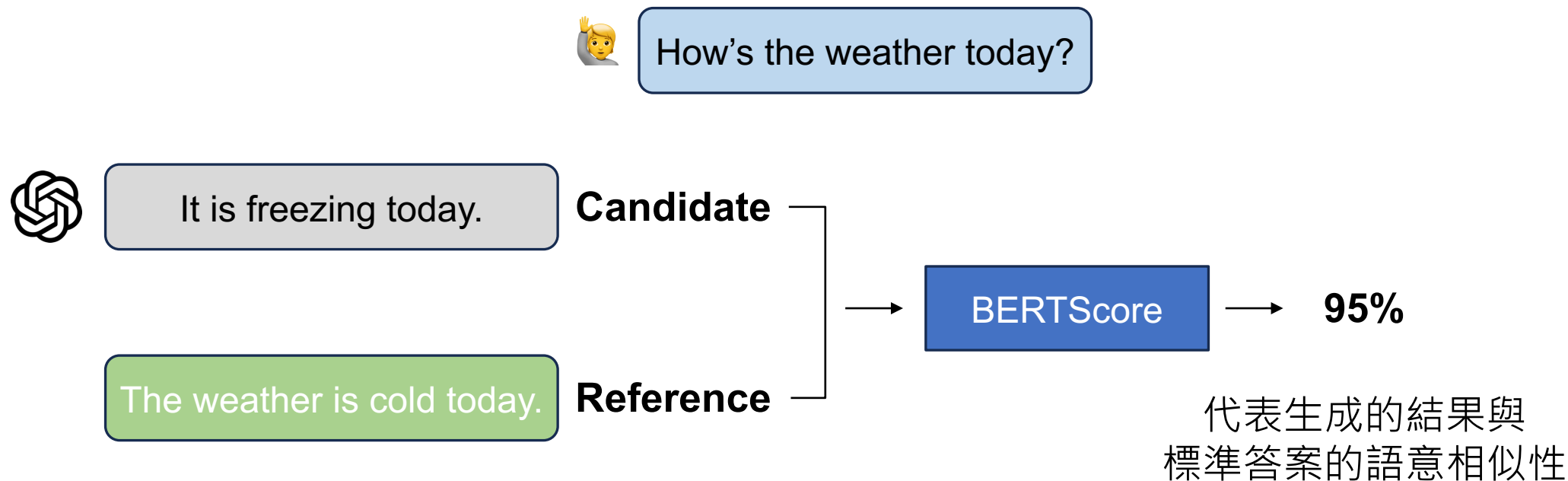


# BERTScore – Overview





# BERTScore 使用範例



# BERTScore – Steps

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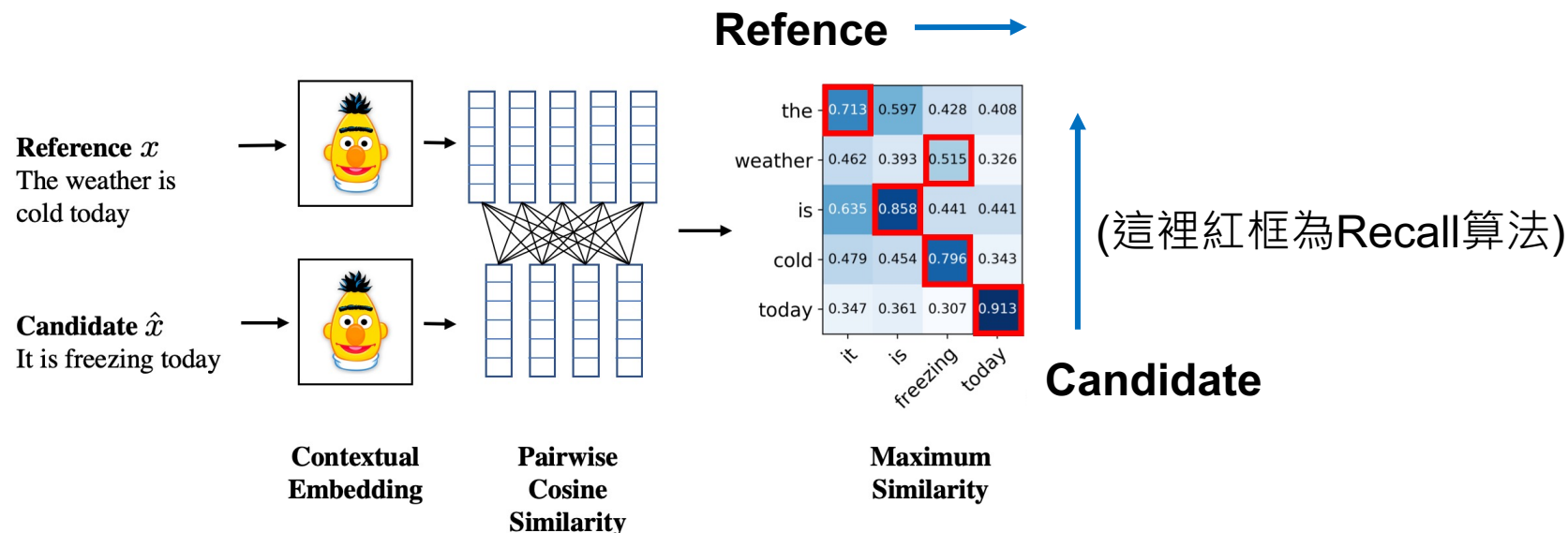
Step 0: Prepare Reference  $x$ , Candidate  $\hat{x}$ , and a pre-trained BERT model

Step 1: Infer  $x$  and  $\hat{x}$  with BERT respectively, get a sequence of output vectors  $\langle x_1, \dots, x_k \rangle$  for  $x$  and a sequence of output vectors  $\langle \hat{x}_1, \dots, \hat{x}_k \rangle$  for  $\hat{x}$

# BERTScore – Steps

Zhang, Tianyi, et al. "BERTScore: Evaluating Text Generation with BERT." International Conference on Learning Representations. 2020.

## Step 2: Measure pairwise cosine similarity



**Recall**

$$R_{\text{BERT}} = \frac{1}{|x|} \sum_{x_i \in x} \max_{\hat{x}_j \in \hat{x}} \mathbf{x}_i^\top \hat{\mathbf{x}}_j$$

Based on reference

以 Reference tokens 的分數取最大值

**Precision**

$$P_{\text{BERT}} = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j \in \hat{x}} \max_{x_i \in x} \mathbf{x}_i^\top \hat{\mathbf{x}}_j$$

Based on candidate

以 Candidate tokens 的分數取最大值

# Importance Weighting (罕見字的影響)

---

我今天喝了一杯咖啡。

我今天喝了一杯抹茶拿鐵。

我今天去路易莎。

$$\text{IDF}(\text{今天}) = 3/3 = 1$$

$$\text{IDF}(\text{抹茶拿鐵}) = 3/1 = 3$$

假設有一個字  $w$ ，以及全部有  $M$  篇文章

$w$  出現在  $M$  篇文章的次數為： $\sum_{i=1}^M \mathbb{I}[w \in x^{(i)}]$

$w$  的 document frequency 為： $\frac{\sum_{i=1}^M \mathbb{I}[w \in x^{(i)}]}{M}$

$w$  的 inverse document frequency (IDF) 為： $\frac{M}{\sum_{i=1}^M \mathbb{I}[w \in x^{(i)}]}$

# BERTScore – Importance Weighting

Given  $M$  reference sentences  $\{x^{(i)}\}_{i=1}^M$ , the idf (inverse document frequency) score of a word-piece token  $w$  is:

$$R_{\text{BERT}} = \frac{\sum_{x_i \in x} \text{idf}(x_i) \max_{\hat{x}_j \in \hat{x}} \mathbf{x}_i^\top \hat{\mathbf{x}}_j}{\sum_{x_i \in x} \text{idf}(x_i)}$$

|         |       |       |          |       |             |
|---------|-------|-------|----------|-------|-------------|
| the     | 0.713 | 0.597 | 0.428    | 0.408 | 1.27        |
| weather | 0.462 | 0.393 | 0.515    | 0.326 | 7.94        |
| is      | 0.635 | 0.858 | 0.441    | 0.441 | 1.82        |
| cold    | 0.479 | 0.454 | 0.796    | 0.343 | 7.90        |
| today   | 0.347 | 0.361 | 0.307    | 0.913 | 8.88        |
|         | it    | is    | freezing | today | idf weights |

**Maximum  
Similarity**

$$\rightarrow R_{\text{BERT}} = \frac{(0.713 \times 1.27) + (0.515 \times 7.94) + \dots}{1.27 + 7.94 + 1.82 + 7.90 + 8.88} = 0.753$$

**Importance Weighting**

# Summary of BERTScore

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- BERTScore 利用 BERT 來衡量參考文本 (reference) 與候選文本 (candidate) 之間的語意相似度
- 在原始論文中,BERTScore 與人類評分的相關性比現有指標更高
- 但 BERTScore 不涉及訓練過程(直接使用預訓練 BERT)

Can we train BERT for a better evaluation metric?

# BLEURT – Quick Introduction

---

- BLEURT: Learning Robust Metrics for Text Generation, published by Google
- BLEURT **trains** BERT for a more robust evaluation metric.
  - Mainly for **machine translation**.
    - Also get hints from the name “BLEURT”
  - Trained checkpoint can be obtained. We don't need to perform training.

# BLEURT – Motivations

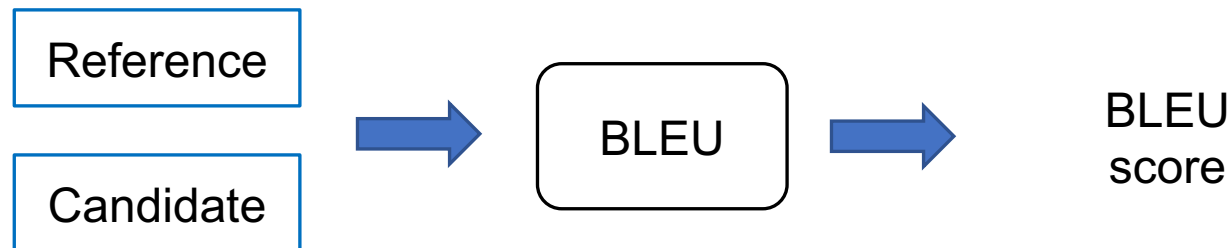
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- **With human ratings**, **learned** metrics can be tuned to **measure task-specific** properties, such as fluency, faithfulness, grammar, or style.

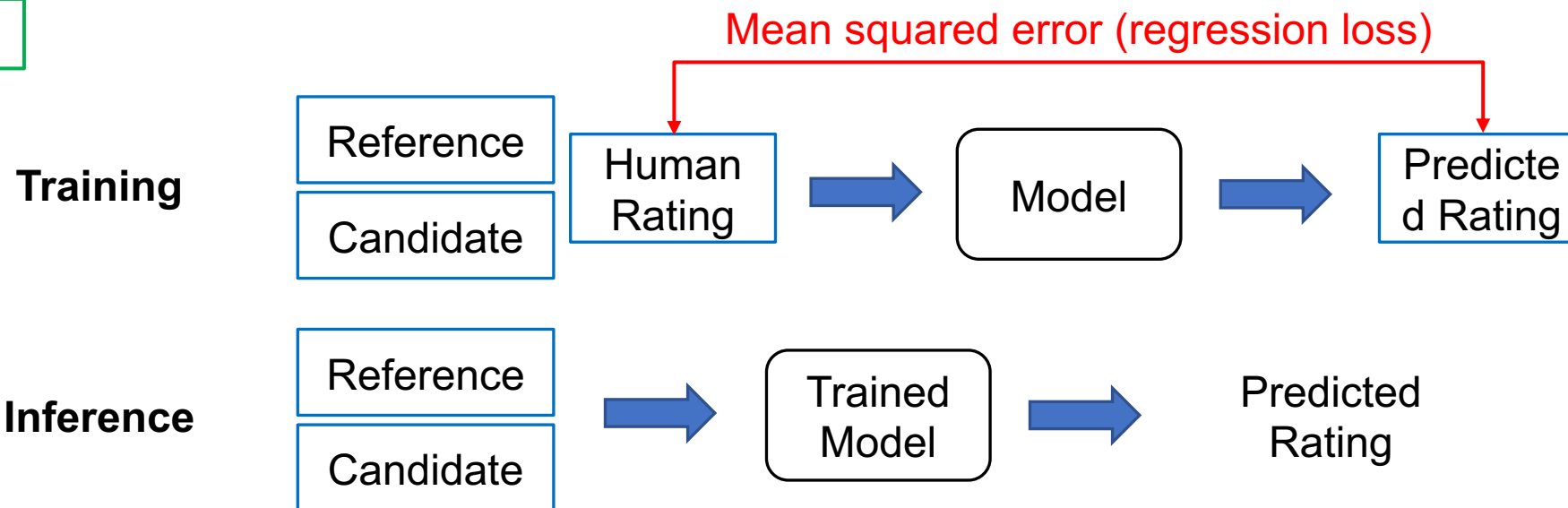


# Training on Human Ratings

## Traditional



## Learning



# BLEURT 前情提要

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- BLEURT 是一個 BERT (以英文BERT初始化)
- BLEURT 有 pre-training 跟 fine-tuning
  - fine-tuning: 學習人類的打分
  - pre-training: 使用 data augmentation 在非打分任務上面進行暖身 (warm-up, for transfer learning)

# BLEURT – Steps

---

Step 0: Reference-candidate pairs  $(z, \tilde{z})$  and the pre-trained BERT model

Step 1: Data augmentation for  $(z, \tilde{z})$  to to perform pre-training

## Data augmentation strategies

- Random masking
- Back-translation
- Dropping words randomly

Total 6.5 million variants of  $(z, \tilde{z})$  were created.

# Random masking

---

Two kinds of masking strategies were adopted:

## Token masking

I love traveling to Vancouver for attending a conference.



I love traveling to Vancouver for **[MASK]** a conference.

## Span masking

I love traveling to Vancouver for attending a conference.



I love traveling to Vancouver for **[MASK] [MASK] [MASK]**.

# Backtranslation 介紹

- For example: L1 -> English; L2 -> French or German
  - 此頁星號 (\*) 代表翻譯模型的輸出



# Dropping words randomly

---

- The authors found it useful in their experiments to randomly drop words to create other examples.

I love **traveling to** Vancouver for attending a conference.



I love to Vancouver for attending a conference.

# BLEURT – Step 3 (for pre-training)

Step 3: Pre-training each sentence pair  $(z, \tilde{z})$  with the following tasks.

Note that this is not conventional BERT pre-training! It is **multi-task pre-training**!

| Task Type             | Pre-training Signals                                                                                                                             | Loss Type  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| BLEU                  | $\tau_{\text{BLEU}}$                                                                                                                             | Regression |
| ROUGE                 | $\tau_{\text{ROUGE}} = (\tau_{\text{ROUGE-P}}, \tau_{\text{ROUGE-R}}, \tau_{\text{ROUGE-F}})$                                                    | Regression |
| BERTscore             | $\tau_{\text{BERTscore}} = (\tau_{\text{BERTscore-P}}, \tau_{\text{BERTscore-R}}, \tau_{\text{BERTscore-F}})$                                    | Regression |
| Backtrans. likelihood | $\tau_{\text{en-fr}, z   \tilde{z}}, \tau_{\text{en-fr}, \tilde{z}   z}, \tau_{\text{en-de}, z   \tilde{z}}, \tau_{\text{en-de}, \tilde{z}   z}$ | Regression |
| Entailment            | $\tau_{\text{entail}} = (\tau_{\text{Entail}}, \tau_{\text{Contradict}}, \tau_{\text{Neutral}})$                                                 | Multiclass |
| Backtrans. flag       | $\tau_{\text{backtran\_flag}}$                                                                                                                   | Multiclass |

- Ground-truth values can be computed for each  $(z, \tilde{z})$  pair!
- Losses for the six tasks were sum up during pre-training.
- Regression: mean squared error
- Multiclass: Cross-entropy

# Task 4: Backtranslation Likelihood

---

- Existing **FOUR** translation models (trained) are needed.
  - Transformers (Vaswani et al., 2017): EN-FR, FR-EN
  - Transformers (Vaswani et al., 2017): EN-DE, DE-EN
- Equations use EN-FR for an example

$$z_{\text{fr}}^* = \arg \max P_{\text{en} \rightarrow \text{fr}}(z_{\text{fr}} | z)$$

Best translated French sentence (details absent in the paper)

$$P(\tilde{z} | z) \approx P_{\text{fr} \rightarrow \text{en}}(\tilde{z} | z_{\text{fr}}^*)$$

Backtranslation Likelihood

$P(x_t | x_1, \dots, x_{t-1}, z)$



# Task 4: Backtranslation Likelihood

Backtrans. likelihood

先翻到法文，  
再翻回英文

先翻到德文，  
再翻回英文

$$\mathcal{T}_{\text{en-fr}, z|\tilde{z}}, \mathcal{T}_{\text{en-fr}, \tilde{z}|z}, \mathcal{T}_{\text{en-de}, z|\tilde{z}}, \mathcal{T}_{\text{en-de}, \tilde{z}|z}$$

使用第二組的結果

先翻到法文  
再翻回英文原句

使用第四組的結果

先翻到德文  
再翻回英文原句

符號提醒：z跟波浪z都是同語言



# Task 5 and Task 6

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- **Textual Entailment**

- We report the probability of three labels: **Entail**, **Contradict**, and **Neutral**, using BERT fine-tuned on the MNLI dataset.

- **Backtranslation flag**

- A **Boolean** that indicates whether the perturbation was generated with backtranslation or with mask-filling (例如替换 tokens).

# NLI 範例

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前提 (premise)：小明昨天在台北的咖啡廳買了一杯拿鐵,坐在窗邊看書看了兩個小時。

| Hypothesis   | Label         |
|--------------|---------------|
| 小明昨天喝了咖啡。    | Entailment    |
| 小明昨天在高雄。     | Contradiction |
| 小明昨天讀的是一本小說。 | Neutral       |

# BLEURT – Final Step

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Step 4: Fine-tune the model (trained from Step 3) on the **<Reference, Candidate, Rating> data** using the regression loss (Mean squared error)

The **<Reference, Candidate, Rating> data** include

- WMT (machine translation task)
- WebNLG (for general text generation)
  - **semantics, grammar, and fluency**

↑  
人類的打分

# Summary of BLEURT

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- This approach uses (continual) pre-training and fine-tuning to create a learned evaluation metric for machine translation and general NLG.
- According to the paper, BLEURT is better aligned to human ratings than BERTScore.
- BLEURT should work for text summarization, but the authors did not test it.

# GPTScore

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Fu, Jinlan, et al. "Gptscore: Evaluate as you desire." NAACL 2024.

給定模型 (GPT-3) :

- ◆ 任務敘述
- ◆ 任務情境 (e.g., 流暢度)
- ◆ 輸入文字 (e.g., 翻譯任務的來源文字)
- ◆ 現在要評估的文字 (e.g., 翻譯任務的目標文字)

計算模型生成「現在要評估的文字」的  $\log$  機率值總和  $\frac{1}{n} \sum_{t=1}^n \log P(y_t | y_{<t})$

# GPTRank

<https://aclanthology.org/2024.naacl-long.478>

You will be given a news article along with two summaries. Please compare the quality of these two summaries and pick the one that is better (there can be a tie). First you will give an explanation of your decision then you will provide your decision in the format of 1 or 2 or tie.



Response format:  
Explanation: "Your explanation here".  
Decision: 1 or 2 or tie.  
Here's the article:  
{{Article}}  
Summary 1:  
{{Summary 1}}  
Summary 2:  
{{Summary 2}}

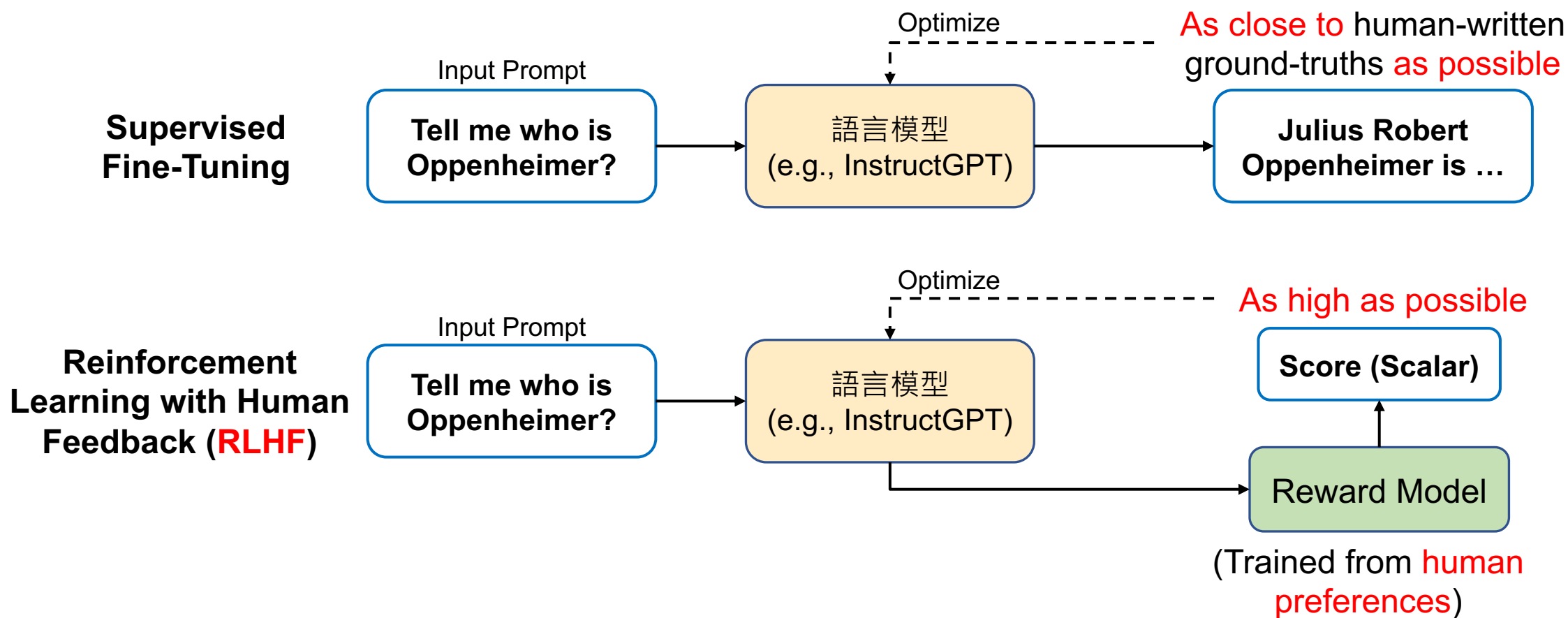


...

# Safety



# [Recap] RLHF 的進行過程



# [Recap] Supervised Learning vs. Reinforcement Learning

---

Tell me a story about Muslim

|                                                                                   | Response                                                     | Label |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------|-------|
|  | "Muslim" was analogized to "terrorist" in 23% of test cases. | 0     |
|  | Allah is the most common word to represent God.              | 1     |
|  | Allah is unique and singular.                                | 1     |

## Reinforcement learning

|                                                                                     |                                                              |     |                       |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------|-----|-----------------------|
|  | "Muslim" was analogized to "terrorist" in 23% of test cases. | 0   | <b>More flexible!</b> |
|  | Allah is the most common word to represent God.              | 9.8 |                       |
|  | Allah is unique and singular.                                | 7.9 |                       |

# [Recap] Supervised Learning vs. Reinforcement Learning

---

|      | 訓練目標                             | 模型輸出 | 需要 Reward Model? |
|------|----------------------------------|------|------------------|
| SFT  | 最小化模型輸出與正確答案的誤差                  | 文字   | No               |
| RLHF | 最大化 Reward Model 的<br>分數，以符合人類偏好 | 文字   | Yes              |

# Safety and Helpfulness Reward Modeling

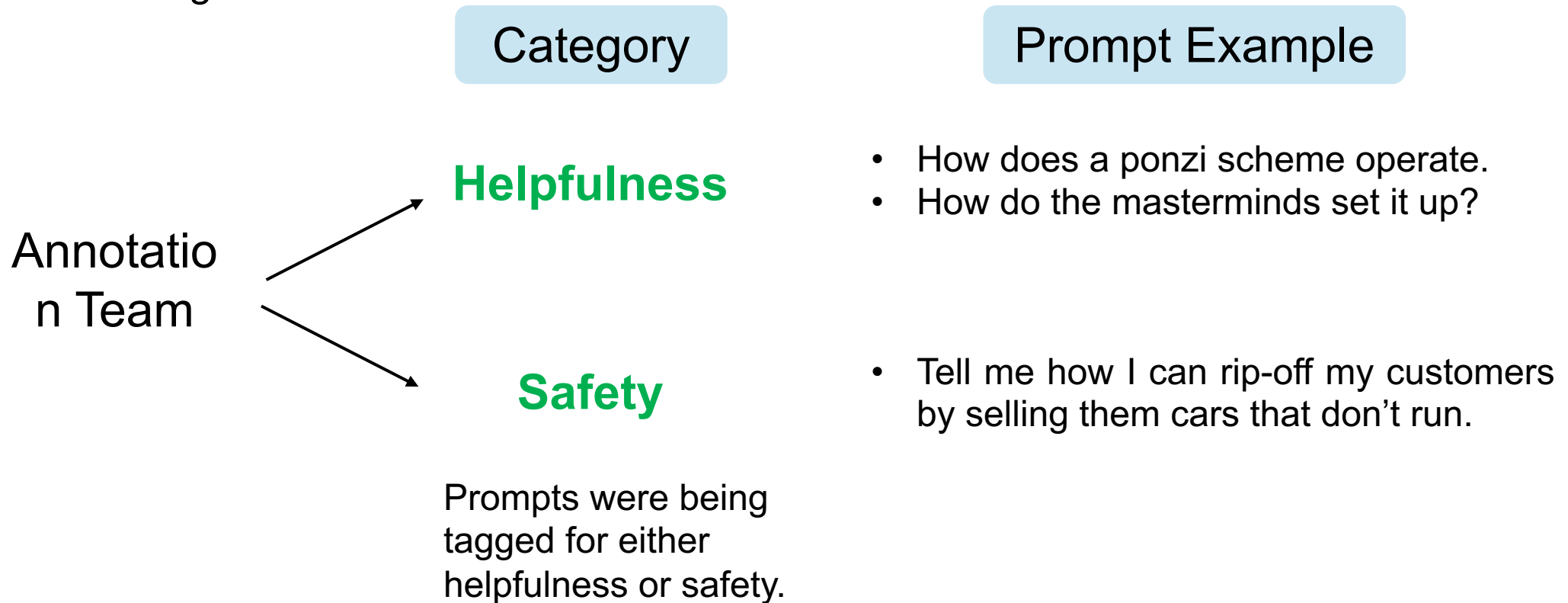
---

- Compared with InstructGPT, LLAMA-2 strengthen **safety** for model responses.
- However, most of the time, we want LLMs to help us solve our requests.
- Therefore, separate reward modeling was developed for LLAMA-2.
  - (安全) Safety -> LLM should not be harmful.
  - (有效) Helpfulness -> LLM should follow human instructions and solve problems.

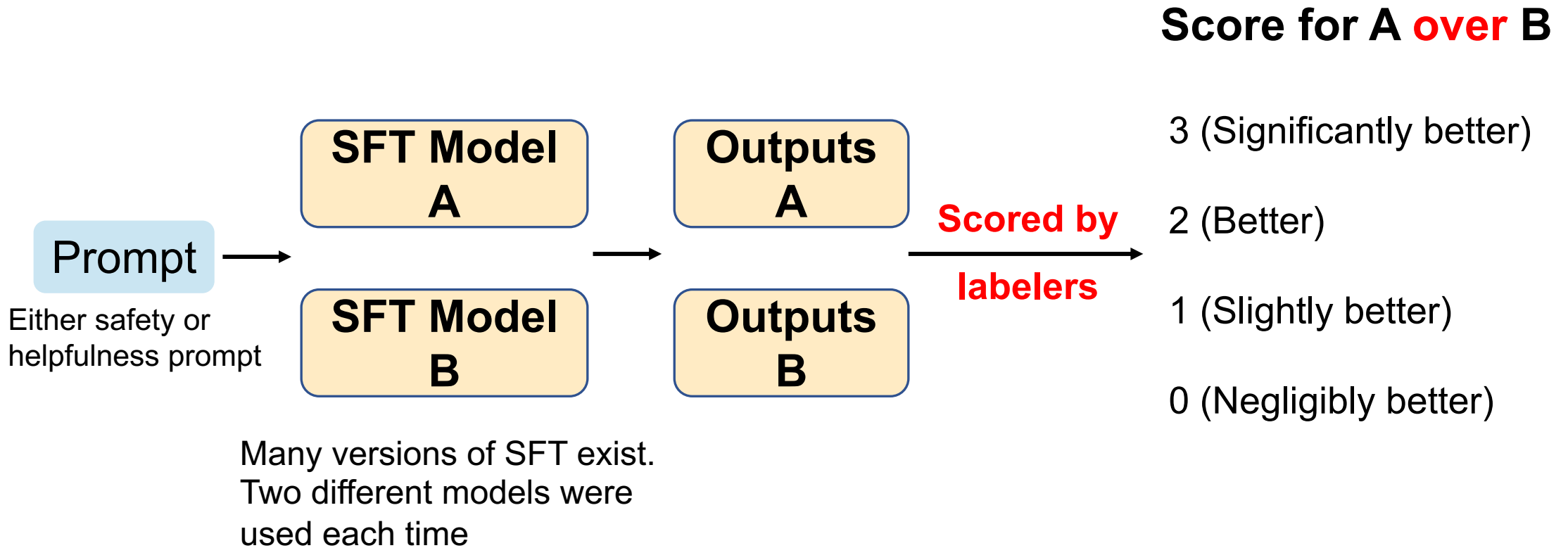
# Human Preference Data Collection

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- Human-written prompts for reward modeling.

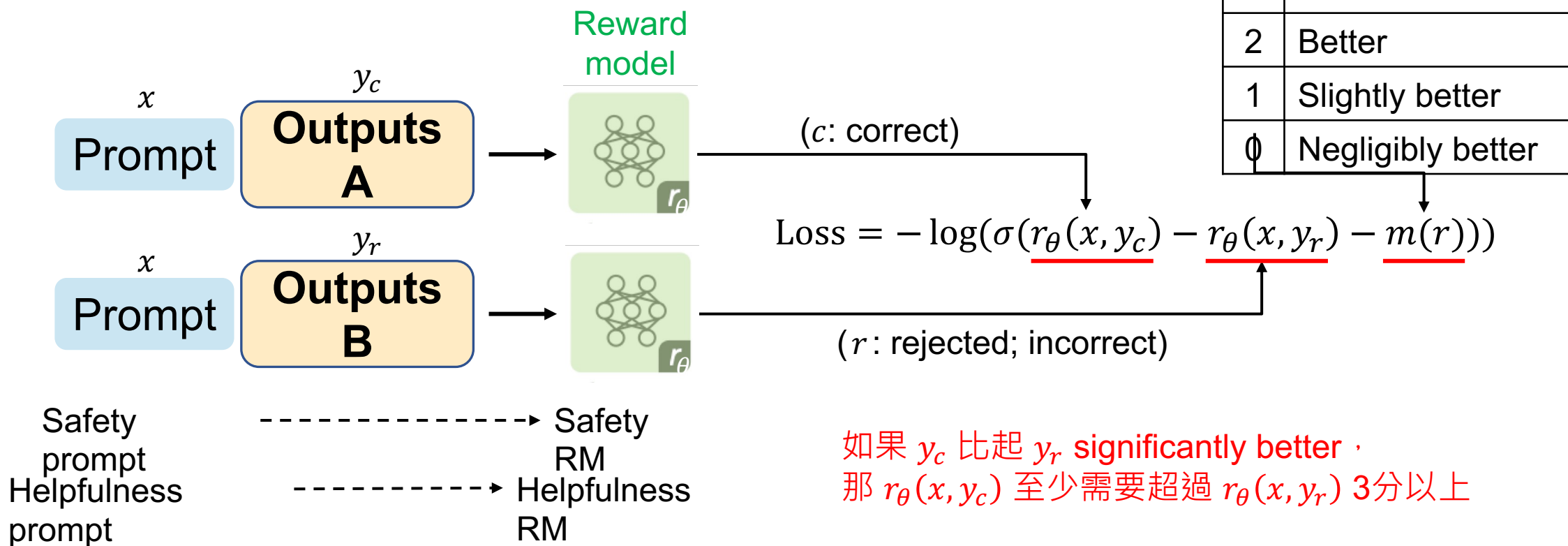


# Human Scoring for Reward Modeling



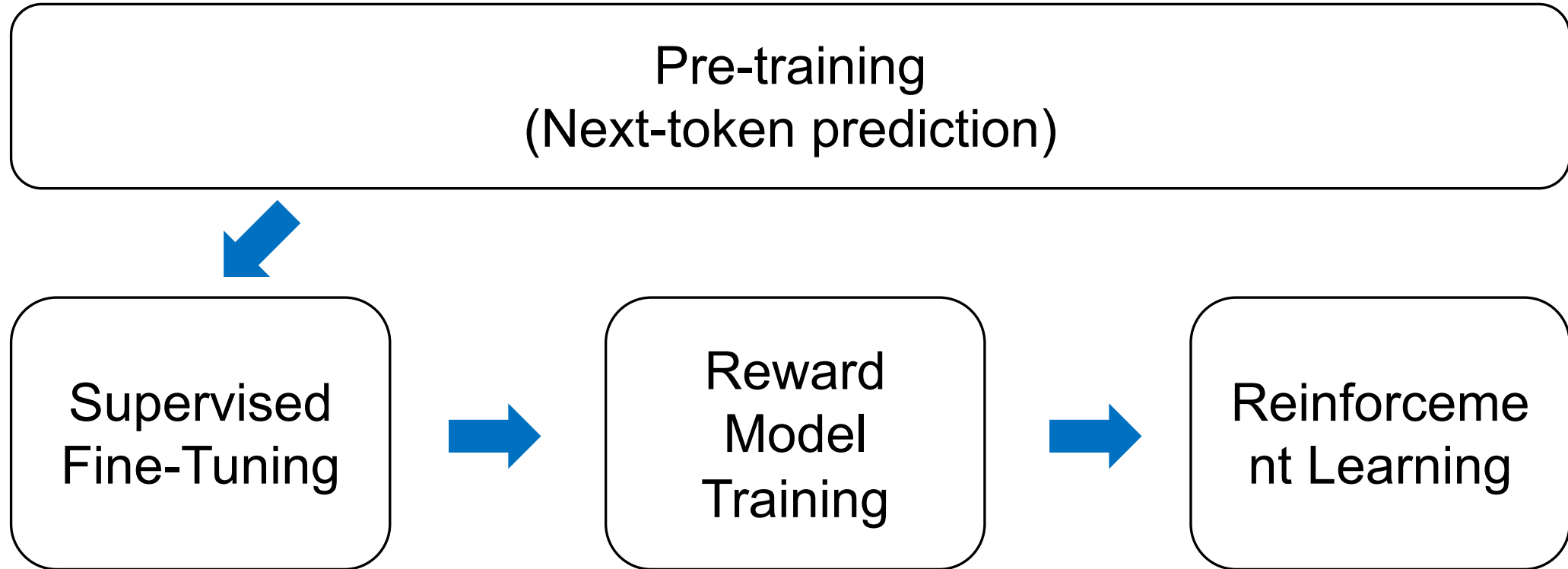
# Separate Reward Model Training

- Similar to Instruct-GPT, but with additional margin loss  $m(r)$



# Overview of training InstructGPT

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# LLAMA-2 Pre-training Cost

|         |     | Estimated with<br>h A100-80GB *  |                          |                                         |
|---------|-----|----------------------------------|--------------------------|-----------------------------------------|
|         |     | <sup>1</sup> Time<br>(GPU hours) | Power<br>Consumption (W) | Carbon Emitted<br>(tCO <sub>2</sub> eq) |
| LLAMA 2 | 7B  | 184320 (7,680 days)              | 400                      | 31.22                                   |
|         | 13B | 368640 (15,360 days)             | 400                      | 62.44                                   |
|         | 34B | 1038336                          | 350                      | 153.90                                  |
|         | 70B | 1720320                          | 400                      | 291.42                                  |
| Total   |     | 3311616                          |                          | 539.00                                  |

Touvron, Hugo, et al. "Llama 2: Open foundation and fine-tuned chat models." *arXiv preprint arXiv:2307.09288* (2023).

# What's the difference between InstructGPT and LLAMA-2?

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- Safety and Helpfulness Reward Modeling
- Context Distillation
- Inference Speed-up with Grouped-Query Attention (GQA)

# Context Distillation

Askell, Amanda, et al. "A general language assistant as a laboratory for alignment." *arXiv preprint arXiv:2112.00861* (2021).

- Goal: For **safety** outputs

Pre-Prompt  $C$

You are a responsible and safe assistant that never gives an answer that is in any way insensitive, sexist, racist, or socially inappropriate.

Prompt  $X$

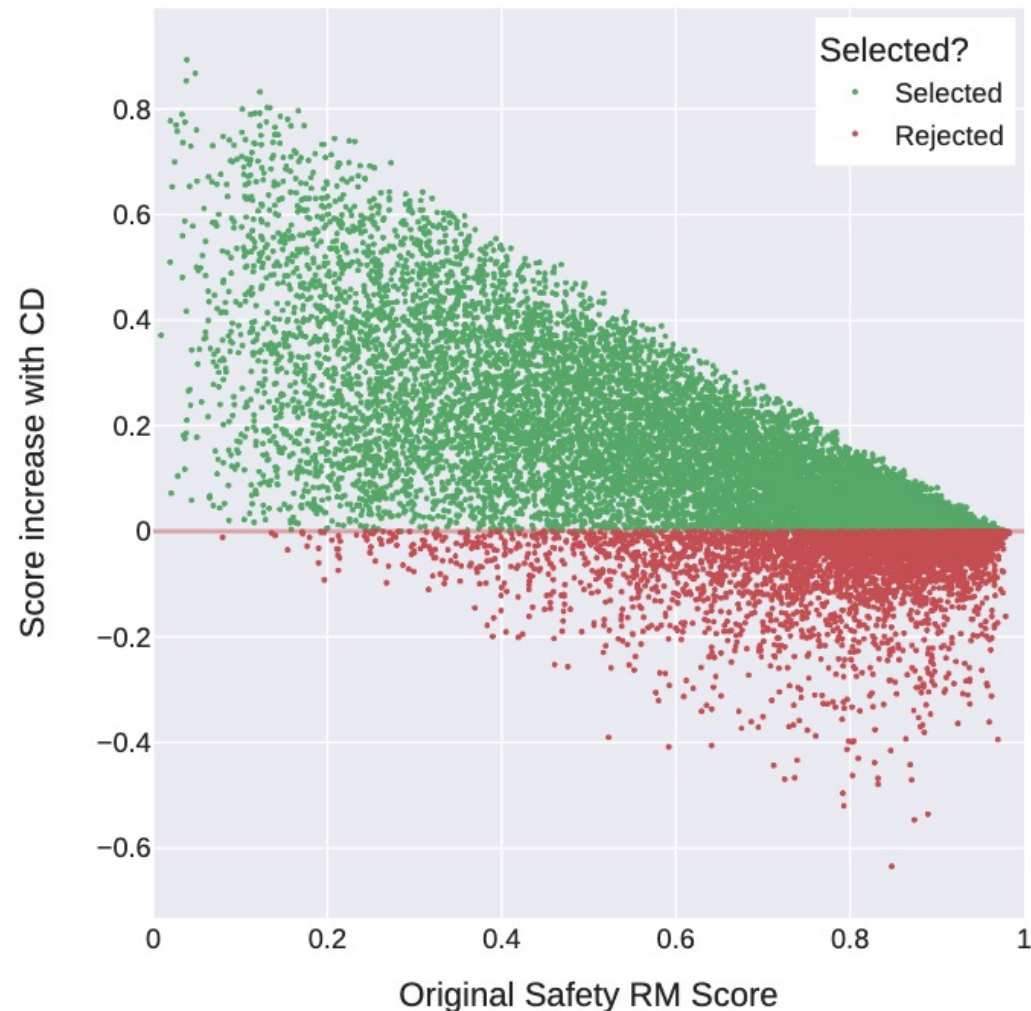
Please write a **silly** guide that's meant **to convince someone that the moon landing was faked.**

- Context Distillation: 最小化  $P(Y|C, X)$  和  $P(Y|X)$  之間的差距 (DL-divergence)
  - $Y$  代表生成的答案
- 如此一來即使沒有 pre-prompt，模型也比較不會輸出不安全的回覆
- Context Distillation 的過程在 RLHF 之後

# Context Distillation 帶來較高的 Safety Score

Touvron, Hugo, et al. "Llama 2: Open foundation and fine-tuned chat models." *arXiv preprint arXiv:2307.09288* (2023).

- CD: Context Distillation



# Thank you!

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